Prateek Kumar Sahu

PhD Candidate
Department of Psychology
Univeristy of Alberta, CA
Proficiency in coding languages: **R** (Expert), Python (Moderate)
Proficient in Causal inference, **Bayesian** modeling, **Bio-acoustics**

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Linkedin, GitHub

Education

- **University of Alberta** — Edmonton, Canada
  PhD, Department of Psychology — *2022 - Now*
- **University of Alberta** — Edmonton, Canada
  MSc, Department of Psychology — *2019 - 2022*
- **Indian Institute of Science Education and Research** — Kolkata, India
  BSc & MSc, Department of Biological Sciences — *2013 - 2018*

Research Experience

- Animal cognition, Vocal production and perception, Novel bioacoustic techniques, open-source research hardware design, preferences in songbirds — Doctoral project
  Dr.Christopher Sturdy — *2022-now*
- Methodological comparisons of classification techniques and refinement of an operant go/no-go procedure — Masters' thesis
  Dr.Christopher Sturdy — *2019-2022*
- Neural basis of sugar-elicited search behaviour of Drosophila — Research Project
  Dr.Axel Brockmann — *2018-2019*
- Understanding reunification dynamics in ant colonies — Summer Project, Masters' thesis
  Dr.Sumana Annagiri — *2015,2017-2018*
- Consistency in personality traits of the shoals of climbing perch — Summer Project
  Dr.V. V. Binoy — *2016*

Publications

14. ***Sahu, P.K.***,*Garcia, A.D.,Deimeke, M.J., Smeltz, S.M.L., Stenstrom, K.H.,Curiston, P., Okocha O.,& Sturdy, C.B.* (2026) Vocal consistency is associated with auditory perceptual ability in a songbird. iScience, 29 (9), https://doi.org/10.1016/j.isci.2026.117219

13. ***Sahu, P.K.***,*Nottebrock, K., Ratch, J., Smeltz, S.M.L., Stenstrom, K.H., Deimeke, M.J.,& Sturdy, C.B.* (2026) Differential response variability of black-capped chickadees to wingbeat sounds and vocalizations. Biology Open, 15(6), bio062598, https://doi.org/10.1242/bio.062598

12. *Stenstrom, K.H., Deimeke, M.J.,* ***Sahu, P.K.****, Smeltz, S.M.L., & Sturdy, C.B.* (2026). What is rescue anyways: A comprehensive review of rescue and rescue adjacent behavior across taxa. Animal Behavior and Cognition, 13(2), 197-213. https://doi.org/10.26451/abc.13.02.06.2026

11. *Stenstrom, K.H., Deimeke, M.J.,* ***Sahu, P.K.****, Smeltz, S.M.L., & Sturdy, C.B.* (2026). Risk versus reward: A systematic review on the threat mobbers face. Animal Behavior and Cognition, 13(2), 184-196. https://doi.org/10.26451/abc.13.02.05.2026

10. ***Sahu, P.K.****, Deimeke, M.J., Garcia, A.D., Stenstrom, K.H., Smeltz, S.M.L., Wang, Y., Ratch, J., & Sturdy, C.B.* (2026) Re-examining hidden fitness: Female preferences for long-path songs in zebra finches. PLOS ONE, 21(3):e0343886, https://doi.org/10.1371/journal.pone.0343886

9. ***Sahu, P.K.****, Garcia, A.D., Stenstrom, K.H., Deimeke, M.J., Smeltz, S.M.L., Lank, I.J.,& Sturdy, C.B.* (2025) A cost-effective Raspberry pi-based operant playback setup to evaluate auditory preferences in songbirds. Behavioural Processes, 232, 105276, https://doi.org/10.1016/j.beproc.2025.105276

8. *Smeltz, S.M.L., Deimeke, M.J., Montenegro, C.,* ***Sahu, P.K.****, Stenstrom, K.H., Camacho-Alpizar, A., & Sturdy, C.B.* (2025) Individual discrimination within, but not between, two vocalization types of the black-capped chickadee. Ethology, e13572. https://doi.org/10.1111/eth.13572

7. *Smeltz, S.M.L., Deimeke, M.J.,* ***Sahu, P.K.****, Montenegro, C., Stenstrom, K.H., Starenchak, I., Rennie, V., Ebrahim, I.M., Jones, J.A., & Sturdy, C.B.* (2025) Individual vocal identity may be obscured following colony assembly in captive black-capped chickadees. Behavioural Processes, 225, 105151, https://doi.org/10.1016/j.beproc.2025.105151

6. ***Sahu, P.K.****, Montenegro, C., Lambert, C.T., Oprea, A., Deimeke, M.J., Rennie, V., Smeltz, S.M.L, Benowicz, T.J., Patel, D., Phillmore, L.S., & Sturdy, C.B.* (2022) Effect of feed-time duration on discrimination of vocalizations in a go/no-go operant paradigm. Behavioural Processes, 203, 10477, https://doi.org/10.1016/j.beproc.2022.104777

5. ***Sahu, P.K.****, Campbell, K.A., Oprea, A., Phillmore, L.S., & Sturdy, C.B.* (2022) Comparing methodologies for classification of zebra finch distance calls. The Journal of the Acoustical Society of America, 151, 3305, https://doi.org/10.1121/10.0011401

4. *Lambert, C.T.,* ***Sahu, P.K.****, Sturdy, C.B., & Guillette, L.M.* (2022) Among-individual differences in auditory and physical cognitive abilities in zebra finches. Learning & Behaviour, 50(3), 389-404, https://doi.org/10.3758/s13420-022-00520-w

3. *Montenegro, C.,* ***Sahu, P.K.****,& Sturdy, C.B.*(2021). Individual differences in female black-capped chickadee (*Poecile atricapillus*) fee-bee songs. The Journal of the Acoustical Society of America, 150(4), 3038-3046, https://doi.org/10.1121/10.0006532

2. *Montenegro, C., Service, W.D., Scully, E.N., Mischler, S.K.,* ***Sahu, P.K.****, Benowicz, T.J., Fox, K.V. R.,& Sturdy, C.B.*(2021). The impact of anthropogenic noise on individual identification via female song in Black-capped chickadees (*Poecile atricapillus*). Scientific Reports, 11(1), https://doi.org/10.1038/s41598-021-96504-3

1. ***Sahu, P.K.****, Kolay, S. & Annagiri, S.*(2019) To reunite or not: A study of artificially fragmented *Diacamma indicum* ant colonies. Behavioural Processes 158, 4-10, https://doi.org/10.1016/j.beproc.2018.10.017

Teaching Experience & Service

- Teaching Assistant: Basic Psychological Processes (2019, 2020, 2021, 2022), Behaviour Modification (2020, 2021, 2022, 2023, 2024, 2025,2026), Principle of Learning (2020), Behaviour & Genetics (2021), Neuroethology (2022), Honours Seminar: Writing course (2023), Theory in Learning and Comparative Cognition (2024), Social Influence (2024, 2025), Research Methods Cognition (2025), Human Neuropsychology (2025)
- Skype A Scientist: Talks at various schools (Indonesia, New York, Colorado, New York) (2022-2025)
- Secretary/Treasurer of the Graduate Psychology Association: (2023-2026)
- Judge for the Royce Research Conference, University of Alberta, CA: (2021, 2024, 2025)
- Volunteer for organizing the International Conference on Comparative Cognition, USA: (2023, 2025)

Awards & Scholarships

Andrew Stewart Memorial Graduate Prize, CA 2026
Alberta Graduate Excellence Scholarship (AGES), CA 2025-26
Graduate Student Travel Award, CA 2024-25
Comparative Cognition Society- Ron Weisman Travel Award, USA 2025
Alberta Graduate Excellence Scholarship (AGES), CA 2024-25
Alberta Graduate Excellence Scholarship (AGES), CA 2023-24
Scholarship from Department of Psychology, University of Alberta, CA 2019-22
INSPIRE Fellowship, IN 2013-18
National Means cum Merit Scholarship, IN 2008-10

Conferences

- Royce Research Conference 2026
- International Conference On Comparative Cognition 2026
- Royce Research Conference 2025
- International Conference On Comparative Cognition 2025
- The Canadian Society for Brain, Behaviour and Cognitive Science (CSBBCS) 2024
- International Conference On Comparative Cognition 2023
- International Conference On Comparative Cognition 2021
- Animal Behaviour Society Conference 2020
- Virtual Royce Conference 2020
- Insect Multisensory Integration Workshop 2019

- 2nd Bangalore Meeting on Asian Bees 2019
- Preparatory School on Population Genetics and Evolution 2019
- Understanding Behaviour 2019
- Workshop on Cyberspace and Cyber Security 2017
- The Ambassadors of Evolution Workshop 2017
- The 10th Year Celebration of IISER Kolkata Advances in Life Sciences 2017
- The Frontiers in Modern Biology (FIMB) symposium 2015

Manuscript Reviews

1.*Ostrich (2022)*; 2,3.*PeerJ (2022, 2023)*, 4.*Behavioral Ecology and Sociobiology (2022)*, 5.*Animal Cognition (2023)*, 6.*Animal Behaviour (2023)*, 7.*Animal Behaviour (2023)*, 8.*The Journal of the Acoustical Society of America (2023)*, 9.*Journal of Comparative Psychology (2023)*, 10.*PLOS One (2024)*, 11.*IBIS: International Journal of Avian Science (2024)*, 12.*Journal of Experimental Biology (2025)*, 13,14.*Behavioural Processes (2025, 2025)*, 16,17) *Learning and Motivation (2026, 2026)*, 18,19)*Journal of Ornithology (2026, 2026)*, 20)*Peer Community Journal in Evolutionary Biology (2026)*

Skills

Bio-acoustical techniques & animal learning paradigms . . . . . University of Alberta, 2019-now
Handling birds & animal care . . . . . University of Alberta, 2019-now
Raising & handling Drosophila . . . . . National Centre for Biological Sciences, 2018-19
Insect(ants, bees) research & insect biology . . . . . IISER-K,NCBS 2015-19
Fish research & handling . . . . . National Institute of Advanced Studies, 2016
Coding & Statistics . . . . . "R","Python", "Matlab"
Graphics Designing . . . . . "Photoshop", "Illustrator"